

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models the motion of the ball by forming an equation of motion	AO3.1b	M1	$m \frac{d^2x}{dt^2} = (-12.5mx) \times 2$
	Uses SHM equations to form model for displacement	AO3.1a	M1	$\Rightarrow \frac{d^2x}{dt^2} = -25x$ $\therefore x = A\sin(5t)$
	Uses initial condition to find the constant	AO1.1a	M1	$\Rightarrow \dot{x} = 5A\cos(5t)$
	Obtains correct value for constant	AO1.1b	A1	
	Interprets 'their' value to find minimum distance from P	AO3.2a	A1F	when $t = 0$, $\dot{x} = 0.75$ so $0.75 = 5A$ $A = 0.15$ Hence $x = 0.15\sin(5t)$ Max displacement = 0.15 metres from O, when $\sin(5t) = \pm 1$, so minimum distance from P is 0.75 metres
(b)	Identifies a correct limitation of the model for example friction between ball and the surface or damping effect due to air	AO3.5b	B1	It is unlikely that the surface is perfectly smooth so friction will be acting. The ball will be likely to travel a smaller distance before coming to rest and the minimum distance of the ball from P may actually be greater than that calculated in part (a).
	Correctly infers whether the distance is too big or too small based on the limitation they have identified. Accept any well-reasoned inference.	AO2.2b	R1	
				Total 7 marks

Q2.

(a) $T_{AC} = \frac{45}{0.4} (0.6 + x - 0.4) = 22.5 + 112.5x$
M1: Two tensions in terms of x with $\pm$ a constant term.

A1: First correct tension.

M1A1

$$T_{BC} = \frac{45}{0.4} (0.6 - x - 0.4) = 22.5 + 112.5x$$

A1: second correct tension

A1

$$\text{Resultant} = T_{BC} - T_{AC}$$

$$= 22.5 - 112.5x - (22.5 + 112.5x)$$

$$= -225x$$

A1: Tensions summed correctly.

A1

AG

(b) $9 \frac{d^2x}{dt^2} = -225x$

$$\frac{d^2x}{dt^2} = -\frac{225}{9} x = -25x$$

M1: Showing second derivative = -25x.

4

M1

$\therefore$ SHM

A1: Correct deduction about SHM.

A1

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2

(c) $\omega = 5$

M1: Using ω from part (b).

M1

$$\text{Period} = \frac{2\pi}{5}$$

A1: Correct period.

Accept AWRT 1.26.

A1

2

(d) $a = 0.1$

B1: Use of $a = 0.1$

B1

$$v^2 = 5^2 (0.1^2 - 0.05^2)$$

M1: Equation with -0.05^2 or $0.05^2 -$

A1: Correct equation.

M1A1

$$v = 0.433 \text{ m s}^{-1}$$

A1: Correct speed.

A1

4

(e) $x = 0.1 \cos(5t)$

B1F: Seeing $a = 0.1$ of their earlier value.

B1F

B1F: Correct expression using their values.

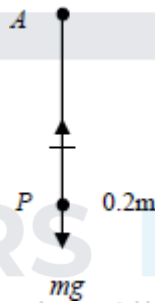
B1F

2

[14]

Q3.

(a)



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$$T = 0.4g$$

$$T = 0.2k$$

M1

$$0.2k = 0.4g$$

A1

$$k = 2g$$

$$k = 19.6 \text{ Nm}^{-1}$$

A1F

Both, accept use of λ and l

3

(b) extension = $0.2 + x$

B1

$$(i) \quad mg - T = m\ddot{x}$$

M1

$$0.4g - 19.6(x + 0.2) = 0.4\ddot{x}$$

ft stiffness

A1F

$$0.4\ddot{x} = -19.6x$$

$$\ddot{x} = -49x \quad (h = -49)$$

A1F

4

ft stiffness

$$(ii) \quad \ddot{x} = -\omega^2 x \quad SHM$$

(constant < 0)

ft stiffness provided h < 0

B1F

1

$$(iii) \quad T = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi}{7}$$

M1

$$T = 0.898 \text{ sec}$$

AG

A1

2

$$(iv) \quad \max v = a\omega$$

$$= 0.1 \times 7$$

M1

$$= 0.7 \text{ ms}^{-1}$$

ft stiffness provided h < 0

A1F

2

[12]

Q4.

$$(a) \quad (i) \quad m\ddot{x} = -\frac{\lambda x}{a}$$

M1

$$= -amn^2 \frac{x}{a}$$

$$\ddot{x} = -n^2 x \quad \text{SHM}$$

A1

2

$$(ii) \quad T = \frac{2\pi}{n}$$

B1

1

$$(b) \quad (i) \quad m\ddot{x} = -amn^2 \frac{x}{a} - mkv$$

3 appropriate terms attempted

Signs consistent

M1

A1

$$\ddot{x} + k\dot{x} + n^2 x = 0$$

AG

A1

3

$$(ii) \quad p^2 + \frac{5n}{2}p + n^2 = 0$$

$$\left(p + \frac{5n}{4}\right)^2 - \frac{9n^2}{16} = 0$$

M1

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$$p + \frac{5n}{4} = \pm \frac{3n}{4}, \quad p = -2n, \quad p = -\frac{n}{2}$$

$$\left. \begin{aligned} 2p^2 + 5np + 2n^2 &= 0 \\ (2p + n)(p + 2n) &= 0 \\ p = -\frac{n}{2}, p &= -2n \end{aligned} \right\}$$

A1

$$x = Ae^{-2nt} + Be^{-\frac{n}{2}t}$$

M1

$$t = 0, x = 0: A + B = 0$$

$$t = 0, \dot{x} = U$$

$$\dot{x} = -2nAe^{-2nt} - \frac{n}{2}Be^{\frac{n}{2}t}$$

m1

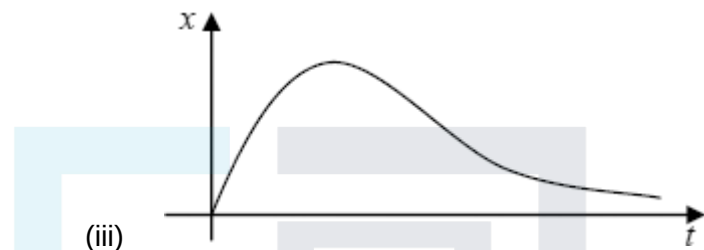
$$U = -2nA - \frac{n}{2}B$$

$$A = -\frac{2U}{3n} \quad B = \frac{2U}{3n}$$

$$x = \frac{2U}{3n} \left(e^{\frac{-nt}{2}} - e^{-2nt} \right)$$

A1,A1

6



(iii)

Accept sketch with correct shape not reaching origin
but not crossing x-axis elsewhere

Accept reference to real distinct roots of auxiliary
equation

B1

Heavy damping

Independent of previous mark

B1

EXAM PAPERS PRACTICE [14]

Q5.

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$$(a) \quad T = 7.5 \Rightarrow 7.5 = \frac{2\pi}{\omega}$$

M1

$$\omega = \frac{4\pi}{15}$$

AG

A1

2

$$(b) \quad (i) \quad \dot{x}^2 = \omega^2(a^2 - x^2) \\ 4 = \omega^2(a^2 - 1)$$

M1

$$a = 2.59 \text{ m}$$

$$AG; a = 2.5883...$$

A1

2

$$(ii) \quad x = a \sin \omega t$$

$$1 = 2.59 \sin \omega t$$

M1

$$\omega t = 0.3964 \text{ (0.3864), } 2.7452 \text{ (2.755)}$$

A1, A1

$$\text{time interval} = (2.7452 - 0.3964) \div \omega$$

m1

$$\text{time} = 2.8 \text{ sec}$$

A1F

5

$$(iii) \quad \text{Max speed} = \omega a$$

$$= 0.8378 \times 2.5883$$

M1

$$= 2.17 \text{ m s}^{-1}$$

A1F

2

EXAM PAPERS PRACTICE [11]

Q6.

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$$(a) \quad x = A \cos 4t + B \sin 4t$$

$$\dot{x} = -4A \sin 4t + 4B \cos 4t$$

B1

$$\ddot{x} = -16A \cos 4t - 16B \sin 4t$$

B1

$$\text{Alt: } m^2 + 16 = 0 \quad B1$$

$$m = \pm 4i \quad B1$$

$$x = A \cos 4t + B \sin 4t \quad M1A1$$

$$\text{Substitute into } \ddot{x} + 16x = 0$$

M1

Satisfactory conclusion

A1

4

(b) $t = 0, \dot{x} = 0: 0 = 0 + 4B \rightarrow B = 0$
AG

B1

1

(c) $x = A \cos 4t$
 $t = \frac{\pi}{12}, x = h: h = A \cos \frac{\pi}{3}$

M1

$$A = 2h$$

A1

2

(d) $\dot{x} = -8h \sin 4t$
Max speed = $8h \text{ m s}^{-1}$

B1

1

(e) $F = m(-32h \cos 4t)$

M1

$$|F_{\max}| = 32hm \text{ N}$$

A1

2

EXAM PAPERS PRACTICE [10]

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Q7.

(a) $2 \times 9.8 = \frac{\lambda}{0.5} \times 0.2$

Equilibrium considered to form equation in λ

M1

$$\lambda = \frac{9.8}{0.2} = 49 \text{ N}$$

Correct equation

A1

Correct λ

A1

3

$$(b) \quad T = \frac{49x}{0.5} + 2g$$

Equation for tension with two terms

M1

Correct equation

A1

$$2 \frac{d^2 x}{dt^2} = 2g - (98x + 2g)$$

$$\text{Use of } F = m \frac{d^2 x}{dt^2}$$

M1

$$\frac{d^2 x}{dt^2} = -\frac{98}{2} x = -49x$$

Correct equation

A1

ag Correct result from correct working

A1

5

$$(c) \quad \text{Period} = \frac{2\pi}{\sqrt{49}}$$

Finding period

M1

Correct period

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A1

$$t = \frac{1}{4} \times \frac{2\pi}{7} = \frac{\pi}{14} = 0.224 \text{ seconds}$$

Dividing period by 4

M1

Correct time

A1

4

[12]

Q8.

$$(a) \quad mg = \frac{\lambda}{l} \times \frac{l}{4}$$

Consideration of forces in equilibrium

M1

$$\lambda = 4mg$$

ag Correct λ from correct working

A1

2

(b) (i) $m \frac{d^2 x}{dt^2} = mg - T$

$$= mg - \frac{4mg}{l} \left(x + \frac{l}{4} \right)$$

Two term expression for T

M1

Correct expression for T

A1

Three term equation of motion

M1

$$= -\frac{4mg}{l} x$$

Simplifying

m1

$$\frac{d^2 x}{dt^2} = -\frac{4g}{l} x$$

ag Correct result from correct working

A1

5

(ii) Period $= 2\pi \sqrt{\frac{1}{4g}} = \pi \sqrt{\frac{1}{g}}$

Identifying ω

M1

Correct period

A1

2

[9]

Q9.

(a) $a = 0.2$

Stating amplitude

B1

$$0.2\omega = 10$$

Using $v = a\omega$

M1

$$\omega = 50$$

Correct value of ω

A1

$$P = \frac{2\pi}{50} = \frac{\pi}{25}$$

Correct period from correct working

A1

4

(b) $v = 50\sqrt{0.2^2 - 0.16^2}$

Using $x = 0.16$ in SHM formula

Correct substitution of all values

M1 A1

$$= 6 \text{ ms}^{-1}$$

Correct speed

A1

3

(c) (i) $\omega = 50, q = 0.2$

Correct ω

Correct q

B1 B1

2

(ii) $0 = p - 0.2 \cos 0$

Using $s = 0$

M1

$$p = 0.2$$

Correct p

A1

2

[11]

Q10.

(a) $a = 0.05$

Amplitude = 0.05

B1

$$6^2 = \omega^2 (0.05^2 - 0.04^2)$$

Use of SHM equation with $x = 0.04$

M1

$$\omega = \sqrt{\frac{0.36}{0.0009}} = 20$$

Correct equation

A1

Solving for ω

m1

Correct ω

A1

$$T = \frac{2\pi}{20} = \frac{\pi}{10}$$

ag Correct period from correct working

A1

6

(b) $v = 0.05 \times 20 = 1 \text{ ms}^{-1}$

Using $v = a\omega$

M1

Correct v

A1

2

(c) $a_{\max} = 0.05 \times 20^2 = 20 \text{ ms}^{-2}$

use of $a_{\max} = a\omega^2$

M1

Correct acceleration

A1

2

Allow ± 20

[10]

Q11.

(a) $mg = \frac{\lambda x}{1}$

Equation for equilibrium

M1

$$e = \frac{mgl}{\lambda}$$

Correct extension

A1

2

$$(b) \quad T = -\frac{\lambda}{1} \left(x + \frac{mgl}{\lambda} \right)$$

Expression for tension

Correct expression

A1

$$m \frac{d^2 x}{dt^2} = mg - T$$

Application of Newton's 2nd law with T and mg

m1

$$= mg - \frac{\lambda}{1} \left(x + \frac{mgl}{\lambda} \right)$$

Simplified to correct final answer

A1

5

$$(c) \quad \omega = 2\pi \sqrt{\frac{\lambda}{ml}}$$

$$P = 2\pi \sqrt{\frac{ml}{\lambda}}$$

Finding period

M1

Correct final answer

A1

2

(d) Increased by a factor of 2

Increased

B1

Factor of 2

B1

2

[11]

Q12.

(a) $t = \frac{1}{4} \times \frac{2\pi}{5}$
 $\omega = 5$
 $\frac{1}{4}$ of their period

B1 M1

$$= \frac{\pi}{10}$$

A1

3

(b) $v_{\max} = 5a$
 $v_{\max} = 5a$

B1

$$\left(\frac{5a}{2}\right)^2 = 5^2(a^2 - x^2)$$

Use of $v^2 = \omega^2(a^2 - x^2)$ with $v = \frac{5a}{2}$

M1

$$x^2 = \frac{3a^2}{4}$$

Correct equation
 Solving for x

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A1 m1

$$x = \frac{a\sqrt{3}}{2}$$

Answer as given or equivalent

A1

5

[8]

Examiner reports

Q2.

In part (a), there were many good responses, but in a number of cases there were issues with the students not obtaining correct expressions for the tensions in the two strings. One

common error was to simply give the tension as $T = \frac{45x}{0.4}$ and take no account of the fact that the string was already stretched. Another common error was to give the expressions

of the form $T = \frac{45x}{0.4} (0.8 \pm x)$ for the tension. Most of these approaches “produced” the printed answer, but credit was only given where this was correctly obtained from correct working. Part (b) was usually done well although some students did not include the mass and ended up with an incorrect value for ω of 15 which they used later in the question.

Part (c) followed on and was also done well in the main. In part (d) several students correctly used the formula $v^2 = \omega^2(a^2 - x)^2$, but a number of students used incorrect values for a . The values 0.5 and 0.7 were seen several times. In part (e), students often gave answers of the correct form, but where they had an incorrect amplitude or value for ω from earlier incorrect working this was carried through into their answers.

Q3.

This was another popular question for many, although a few failed to appreciate the vertical nature of the motion, falling down in part (b)(i). Parts (a) and (b)(ii) and (iii) were mostly successful but a few could not provide a correct formula for the maximum speed for this motion.

Q4.

Candidates were able to attempt all parts of this question, but part (a) proved more challenging than expected, with some seeming unsure as to the requirements for simple harmonic motion. Part (b)(i) was done very well, as was part (b)(ii), although there were a number of errors in accuracy here. Part (b)(iii) proved very challenging; many successfully quoted the nature of the roots of the auxiliary equation as justification for heavy damping, but the hint of using a sketch was not appreciated.

Q5.

This was a popular question, showing sound knowledge of techniques relevant to the topic. Part (a) was done very well. Part (b)(i) was attempted correctly by most candidates but was sometimes marred by inaccurate work. Part (b)(ii) revealed a lack of understanding of the cyclical harmonic motion under consideration, with few candidates giving fully correct solutions. Part (b)(iii) was answered very well; a good source of marks.

Q6.

Again, this question was answered well, with candidates showing good knowledge and understanding of this topic. The appropriate techniques were usually applied efficiently to produce correct solutions.

Q7.

The candidates did well with part (a) and for quite a number of candidates these were the only marks that they obtained on this question.

In part (b), the majority of the candidates totally ignored the weight of the sphere and produced invalid arguments. There were some candidates who could produce the required result without difficulty.

In the last part, there were quite a few candidates who found the period, but did not realise that the time needed was a quarter of the period.

Q8.

Part (a) of this question was done very well by almost all of the candidates. Part (b)(i) was very much more demanding and there were very few correct solutions. Most of the candidates did not find a two term expression for the tension in the string and did not include gravity when applying Newton's second law. In contrast, part (b)(ii) was done quite well, although some candidates did have difficulty simplifying their results.

Q9.

A good number of candidates produced good solutions to part (a). Part (b) was found to be very much more difficult, with candidates substituting the wrong value for x or quoting the formula incorrectly. In part (c)(i), the values were quite frequently given correctly, although some candidates gave a negative value for q . In part (c)(ii), quite a few candidates did not realise that when $t = 0$ s was also zero.

Q10.

This question was generally either done very well or very badly. The candidates who were able to identify the amplitude and use the displacement correctly usually went on to gain full marks. The candidates who did not make these steps made very little progress. Some candidates were able to use the period to enable them to answer parts (b) and (c), even though they had not been able to derive it themselves.

Q11.

There were many good answers to part (a). Virtually all those who realised that they needed to consider when the forces were in equilibrium gained both marks. Part (b) was found to be very much more difficult. Several candidates did not include the weight of the particle at any stage of their solutions. Some others produced very contrived arguments to produce the negative sign in the differential equation. Part (c) was often done well, and those who did not gain marks on this question did not seem to know how to proceed. In part (d), several of the candidates stated that the period would increase, but a smaller number were able to quantify the increase correctly.

Q12.

There were some good solutions to this question, but many candidates found it difficult to make much progress. Part (a) in particular caused a lot of difficulty. Quite a number of candidates did not realise that a quarter of the period was needed, often using a half instead. In part (b), there were more correct solutions, but some candidates did not seem to know how to approach the problem.